

Souita Charaf

Software Engineer | Backend & Technical Content Evaluator

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Profile

A dedicated Software Engineering student at 1337 (42 Network) bridging the gap between low-level systems programming and modern web infrastructure. Leveraging a rigorous technical foundation, I excel at building high-performance server architectures, managing containerized deployments, and writing secure, scalable backend code. I specialize in identifying performance bottlenecks and ensuring architectural consistency across complex environments.

Skills

• Backend & Systems Programming

- TypeScript, JavaScript (ES6+), C++98
- NestJS, RESTful APIs, Socket Programming, CRUD Architectures

Databases & Data Modeling

- PostgreSQL, MariaDB, Prisma ORM

Infrastructure & DevOps

- Unix/Linux System Administration (Debian, Kali, WSL)
- Docker, NGINX, LEMP Stack, Bash Scripting

Industrial IT & Real-Time Systems

- PLC Logic, Signal Processing, Real-Time Activity Feeds

Projects

Multi-Service Containerized Infrastructure

- **Designed and deployed** a complete cloud-native environment using **Docker** and **Docker Compose**, featuring a secure network of interconnected services including Nginx, MariaDB, and WordPress.
- **Architected** a system where each service runs in an isolated, lightweight container, communicating over a strictly defined private virtual network to ensure high availability and security.
- **Automated** environment configuration and deployment workflows using **Bash scripting**, ensuring 100% consistency across different development environments.

Custom Unix Shell Implementation

- **Collaborated with a partner** to architect a functional command-line shell in **C**, replicating complex Unix behaviors including process management and signal handling.
- **Co-designed a modular architecture** by splitting the project into a Lexical Parser (lexer/parser) and an Execution Engine, ensuring clean communication through shared data structures.
- **Engineered robust process control logic** using system calls (fork, execve, waitpid) and managed file descriptor redirections (<, >, >>, |) in coordination with team-led integration tests.
- **Synchronized development via Git**, implementing a rigorous testing suite to identify and resolve race conditions, memory leaks, and edge cases in a shared codebase.

Real-Time IRC Network Server | Team Lead

- **Development:** Architected a high-precision IRC server in **C++** using non-blocking I/O and the RFC 1459 protocol.
- **Logic Auditing:** This unique blend of engineering rigor allows me to evaluate AI-generated networking code with a focus on structural integrity and technical accuracy.

Collaborative CRM Platform

- **Architecture:** Engineered a centralized backend system using **NestJS** and **TypeScript**, prioritizing clear information architecture and modularity.
- **Technical Rigor:** Developed robust data validation logic and secure CRUD operations, ensuring 100% adherence to performance protocols.
- **Alignment Relevant:** Experienced in analyzing and structuring complex information flows and identifying syntax inconsistencies in server-side logic.

Education

Student At 1337 Coding School 	2023 – Present Ben Guerir , Morocco
Advanced curriculum focusing on Algorithms, Systems, and Full-Stack Development in a high-intensity peer-learning environment.	
Specialized Technician Diploma in Industrial Computing and Automation, ISTA1	2021 – 2023 Safi , Morocco
Baccalauréat in Electrical Sciences and Technologies, Lycée Charif Al Idrissi	2019 – 2020 Safi , Morocco

Languages

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| • Arabic | • English | • Français |
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